

# DIGITAL FORENSICS PRACTICALS — COMPLETE BEGINNER GUIDE

I'll explain every practical from absolute basics:

- What the tool does
  - Why we use it
  - Exact buttons to click
  - What happens after clicking
  - What to write in exam/viva
- 

## 1. WHAT IS DIGITAL FORENSICS?

Digital forensics means:

- investigating computers,
- USB drives,
- emails,
- internet activity,
- deleted files,
- network traffic

without changing original evidence.

Example:

Someone copied pirated software into a USB drive.

You investigate:

- what files exist,
  - when they were created,
  - whether deleted files exist,
  - whether internet activity is suspicious.
- 

## MOST IMPORTANT CONCEPT → HASH

A hash is like a fingerprint of a file.

If file changes even slightly:

- hash changes completely.

Example:

Original File → MD5 = ABC123

Modified File → MD5 = XYZ789

So investigators use:

- MD5
- SHA1

to prove evidence was NOT changed.

---

## **2. FTK IMAGER — BEGINNER GUIDE**

### **What is FTK Imager?**

FTK Imager is used to:

- create forensic image,
- view evidence,
- recover deleted files,
- export files,
- calculate hash values.

Think of it like:

“Read-only File Explorer for forensic investigation.”

---

## **PRACTICAL 1 — OPEN EVIDENCE (ADD EVIDENCE ITEM)**

# PURPOSE

View USB drive or image WITHOUT copying it.

---

## STEP-BY-STEP

### Step 1

Open FTK Imager.

You will see:

- Menu bar
  - Empty left panel
  - Empty right panel
- 

### Step 2

Click:

File → Add Evidence Item

---

### Step 3

A window appears asking:

Select Source Type

You will see options:

| Option         | Meaning          |
|----------------|------------------|
| Physical Drive | Entire hard disk |

|               |                           |
|---------------|---------------------------|
| Logical Drive | C:, D:, USB drive         |
| Image File    | Existing .E01 or .dd file |

---

## Step 4

If using USB:  
Select:

Logical Drive

Click:

Next

---

## Step 5

Select your USB drive.

Example:

E:

Click:

Finish

---

## Step 6

Now left side shows folders/files.

Expand folders using:



## Step 7

Click any file.

Bottom section shows:

- Natural view
  - Text view
  - Hex view
- 

## WHAT IS HEX VIEW?

Hex view shows actual binary data.

Example:

4D 5A

means EXE file.

This proves:  
file is actually executable.

---

## EXAM POINT

### Difference

| Add Evidence Item  | Create Disk Image    |
|--------------------|----------------------|
| Only view evidence | Create forensic copy |
| No image created   | Image created        |

---

# PRACTICAL 2 — CREATE FORENSIC IMAGE

## PURPOSE

Create exact copy of evidence.

Original evidence should NEVER be analyzed directly.

---

## STEP-BY-STEP

### Step 1

Click:

File → Create Disk Image

---

### Step 2

Select:

Logical Drive

(if USB)

Click:

Next

---

### Step 3

Choose USB drive.

Click:

Finish

---

## Step 4

New window opens.

Click:

Add

---

## Step 5

Select image format.

Choose:

E01

because:

- compressed,
- stores hash,
- standard forensic format.

Click:

Next

---

## Step 6

Fill case details:

| Field           | Example |
|-----------------|---------|
| Case Number     | CASE001 |
| Examiner        | Atharva |
| Evidence Number | USB01   |

Click:

Next

---

## Step 7

Choose:

- destination folder
- image filename

Set:

Fragment Size = 0

Meaning:  
single file only.

---

## Step 8

Tick:

✓ Verify image after creation

VERY IMPORTANT.

---

## Step 9

Click:

Start

Imaging begins.

---

## Step 10

After completion:

FTK shows:

| Value | Example |
|-------|---------|
|-------|---------|

|     |         |
|-----|---------|
| MD5 | 8A7F... |
|-----|---------|

|      |         |
|------|---------|
| SHA1 | 2B91... |
|------|---------|

SAVE THESE.

---

## WHY IMPORTANT?

These hashes prove:

image was NOT modified.

---

## PRACTICAL 3 — VERIFY IMAGE

### PURPOSE

Check if forensic image changed.

---

# STEPS

## Step 1

Click:

File → Verify Drive/Image

---

## Step 2

Select **.E01**

---

## Step 3

FTK recalculates hash.

---

## Step 4

Compare with original hash.

| Result    | Meaning  |
|-----------|----------|
| MATCH     | Safe     |
| NOT MATCH | Tampered |

---

# PRACTICAL 4 — EXPORT FILE

## PURPOSE

Recover/copy suspicious file.

---

## STEPS

1. Open evidence
2. Navigate to file
3. Right-click file
4. Click:

Export Files

5. Choose folder
  6. Save
- 

## 3. AUTOPSY — BEGINNER GUIDE

### What is Autopsy?

Autopsy is used for:

- deleted file recovery,
- keyword searching,
- browser history,
- email analysis,
- report generation.

Think:

“Advanced forensic investigation software.”

---

## PRACTICAL 1 — CREATE CASE

---

## Step 1

Open Autopsy.

---

## Step 2

Click:

Create New Case

---

## Step 3

Enter:

| Field          | Example |
|----------------|---------|
| Case Name      | Case1   |
| Base Directory | Desktop |
| Examiner       | Atharva |

---

## Step 4

Click:

Finish

Case created.

---

# PRACTICAL 2 — ADD IMAGE

---

## Step 1

Click:

Add Data Source

---

## Step 2

Select:

Disk Image or VM File

---

## Step 3

Browse .E01

---

## Step 4

Click:

Next

---

## Step 5

Select modules:

- ✓ Keyword Search
  - ✓ Recent Activity
  - ✓ Hash Lookup
- 

## Step 6

Click:

Finish

Autopsy starts analysis.

---

## PRACTICAL 3 — FIND DELETED FILES

---

### Step 1

Left side panel:

Go to:

Views → Deleted Files

---

### Step 2

Deleted files show with:

- red X
- 

### Step 3

Right-click file.

Click:

Extract File(s)

---

### Step 4

Choose destination folder.

File recovered.

---

## PRACTICAL 4 — GENERATE REPORT

---

### Step 1

Click:

Generate Report

---

### Step 2

Select:

Excel (XLSX)

---

### Step 3

Click:

Finish

Report generated.

---

## 4. WIRESHARK — BEGINNER GUIDE

### What is Wireshark?

Wireshark captures internet/network packets.

Think:

“CCTV camera for network traffic.”

---

# PRACTICAL 1 — CAPTURE PACKETS

---

## Step 1

Open Wireshark.

---

## Step 2

Double-click:

- Wi-Fi  
OR
  - Ethernet
- 

## Step 3

Packets start appearing.

Every line = one packet.

---

## Step 4

Open browser and visit website.

More packets appear.

---

## Step 5

Click RED square button.

Capture stops.

---

## Step 6

Save file:

File → Save As

Save as:

.pcapng

---

# PRACTICAL 2 — APPLY FILTERS

## PURPOSE

Show only needed packets.

---

## COMMON FILTERS

| Filter | Meaning      |
|--------|--------------|
| http   | HTTP traffic |
| dns    | DNS traffic  |
| tcp    | TCP traffic  |
| udp    | UDP traffic  |

---

# HOW?

## Step 1

Top filter bar:  
type:

http

---

## Step 2

Press Enter.

Now only HTTP packets appear.

---

# FILTER COLOR

| Color | Meaning |
|-------|---------|
|-------|---------|

|       |         |
|-------|---------|
| Green | Correct |
|-------|---------|

|     |              |
|-----|--------------|
| Red | Wrong syntax |
|-----|--------------|

---

# PRACTICAL 3 — FOLLOW TCP STREAM

## PURPOSE

See complete communication.

---

## Step 1

Select packet.

---

## Step 2

Right-click.

---

## Step 3

Click:

Follow → TCP Stream

---

## Step 4

Full conversation appears.

| Color | Meaning |
|-------|---------|
|-------|---------|

|     |        |
|-----|--------|
| Red | Client |
|-----|--------|

|      |        |
|------|--------|
| Blue | Server |
|------|--------|

---

## USED FOR

- usernames,
  - passwords,
  - chats,
  - requests.
-

# PRACTICAL 4 — EXPORT FILES

---

## Step 1

Open capture.

---

## Step 2

Click:

File → Export Objects → HTTP

---

## Step 3

Select file.

---

## Step 4

Save it.

Recovered from network traffic.

---

# 5. SYSINTERNALS TOOLS

---

## PROCESS MONITOR (ProcMon)

### PURPOSE

Monitor:

- file access,
  - registry activity,
  - process activity.
- 

## STEPS

1. Open ProcMon as Admin
  2. Activity starts automatically
  3. Observe events
  4. Apply filters
  5. Save **.PML**
- 

## RAMCAPTURE

### PURPOSE

Capture RAM memory.

---

## STEPS

1. Run as Admin
2. Select location
3. Click:

Capture

4. **.mem** file created
-

# TCPVIEW

## PURPOSE

See active network connections.

---

## STEPS

1. Open TcpView
  2. Observe:
    - Process Name
    - Remote IP
    - Ports
- 

# 6. EMAIL FORENSICS

---

## PURPOSE

Investigate emails.

---

## STEPS

1. Open FTK
2. Create case
3. Add .pst
4. Open:
  - Inbox
  - Sent
  - Deleted Items

5. Read emails
- 

## IMPORTANT EMAIL HEADERS

| Header           | Meaning   |
|------------------|-----------|
| From             | Sender    |
| To               | Receiver  |
| Subject          | Topic     |
| Date             | Sent time |
| X-Originating-IP | Sender IP |

---

## 7. BROWSER HISTORY EXAMINER

### PURPOSE

Analyze browser history.

---

### STEPS

1. Open Browser History Examiner
2. Click:

Capture History

3. Select:
  - Chrome
  - Firefox
4. Click:

Next

---

## IMPORTANT SECTIONS

| Section        | Meaning          |
|----------------|------------------|
| Website Visits | Sites visited    |
| Searches       | Search history   |
| Downloads      | Downloaded files |
| Cookies        | Website cookies  |

---

## GENERATE REPORT

1. Click:

File → Report

2. Select PDF
  3. Save
- 

## MOST IMPORTANT EXAM FLOW

## IF EXAM GIVES USB DRIVE CASE

### Follow THIS EXACT ORDER

#### Step 1

Create forensic image using FTK Imager.

---

## Step 2

Record:

- MD5
  - SHA1
- 

## Step 3

Open image in Autopsy.

---

## Step 4

Check:

Views → Deleted Files

---

## Step 5

Extract suspicious files.

---

## Step 6

Generate report.

---

# MOST IMPORTANT VIVA QUESTIONS

| Question      | Answer                 |
|---------------|------------------------|
| What is hash? | Fingerprint of data    |
| Why use hash? | Integrity verification |

|                                          |                   |
|------------------------------------------|-------------------|
| Which tool captures RAM?                 | RAMCapture        |
| Which tool analyzes packets?             | Wireshark         |
| Which tool recovers deleted files?       | Autopsy           |
| Which format is standard?                | E01               |
| Which feature reconstructs conversation? | Follow TCP Stream |